

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE
ORGANISATION OF HIGHER NATIONAL
DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma-New program – 2025 Session

Spécialité/option : SOFTWARE ENGINEERING (SWE)

Paper : Computer Technology

Duration : 4hours

Credit : 4

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*
- *Where calculations are required, clearly show your working and be chronological in your answer.*

SECTION A: MCQs (40 MARKS)

One (01) Mark for Each Correct Answer. Write Your Answers in Your Booklet

1. How can you open a link in a new browser window?
 - a.
 - b.
 - c.
 - d.
2. In C programming, a string might be:
 - a. A character
 - b. A set of character
 - c. An array of character
 - d. An integer
3. An unsigned char format specifier is:
 - a. %d
 - b. %f
 - c. %c
 - d. %lu
4. Which programming languages are used for web design:
 - a. XML, HTML, Java
 - b. C#, PHP, CSS
 - c. PHP, AJAX, JavaScript
 - d. CSS, C++, HTML
5. Which topology covers the factors like security, robustness and eliminating traffic?
 - a. Bus

- b. Star
 - c. Mesh
 - d. Ring
6. Multipoint topology is atopology?
- a. Ring
 - b. Star
 - c. Bus
 - d. None
7. Many languages, C and Java included, pass all parameters using
- a. Call by reference
 - b. Call by Value
 - c. Return by value
 - d. Return by reference
8. An unsigned log double format specifier is:
- a. %d
 - b. %f
 - c. %c
 - d. %lu
9. An unsigned float format specifier is:
- a. %d
 - b. %f
 - c. %c
 - d. %lu
10. What does CSS stand for?
- a. Colorful Style Sheets
 - b. Computer Style Sheets
 - c. Cascading Style Sheets
 - d. Creative Style Sheets
11. The formatted input functions that work with the character data type is:
- a. Get()
 - b. Getch()
 - c. Getchs()
 - d. Getches()
12. The formatted input functions that work with the character data type is:
- a. Putchs()
 - b. Putch()
 - c. Putche()
 - d. Put()
13. Inside which HTML element do we put the JavaScript?
- a. <js>
 - b. <script>
 - c. <scripting>
 - d. <javascript>plan

14. What is the output of this program:

```
main()
static i=3; printf("%d",i--);
return i>0? main():0;
```

- a. 123
- b. 213
- c. 321
- d. 132

15. What is the output of this program:

```
char *foo()
char result[100];
strcpy(result,"anything is good");
return(result);
void main()
char *j;
j=foo()
printf("%s",j);
```

- a. nothing
- b. error message
- c. anything is good
- d. the program will not run

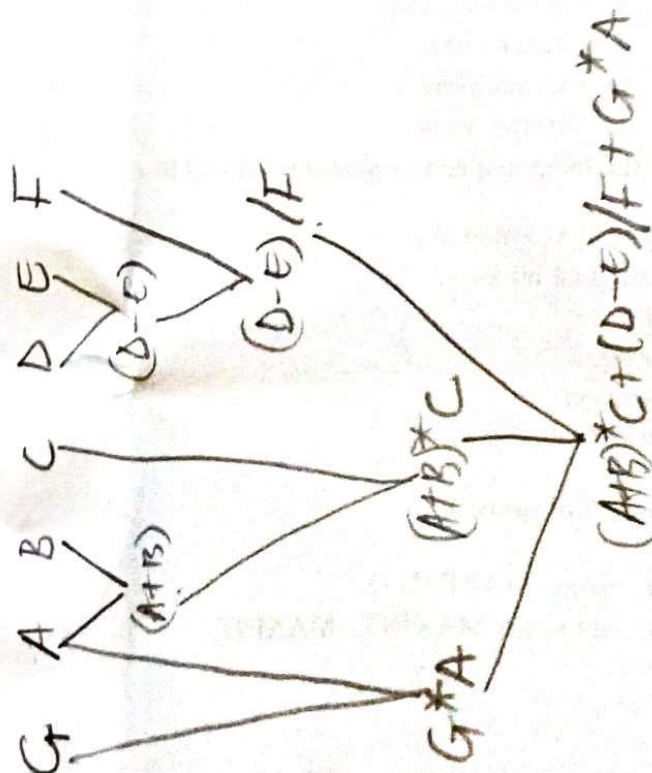
16. One of the steps involves in the Design phase of SDLC is:

- a. design strategy
- b. analysis and initial design
- c. requirements gathering
- d. support plan

17. Output of this program is:

```
main()
int i=0;
for(i=0;i<20;i++)
{switch(i)
case 0:i+=5;
case 1:i+=2;
case 5:i+=5;
default i+=4;
break;}
printf("%d",i);}
```

- a. 0,5,9,13,7
- b. 16,21
- c. 12,17,22
- d. syntax error



18- What does PHP stand for?

- a. Hypertext Preprocessor
- b. Personal Hypertext Processor
- c. Private Home Page
- d. All above

19- What will the following program do:

```
void main()
{ int i;
  char a[]="String";
  char *p="New String";
  char *Temp;
  Temp=a;
  a=malloc(strlen(p) + 1);
  strcpy(a,p); //Line number:9//
  p = malloc(strlen(Temp) + 1);
  strcpy(p,Temp);
  printf("(%s, %s)",a,p);
  free(p);
  free(a);
} //Line number 15//
```

- a. Swap contents of p & a and print:(New string, string)
- b. Generate compilation error in line number 8
- c. Generate compilation error in line number 5
- d. Generate compilation error in line number 7

20- The view of total database content is:

- a. Conceptual view.
- b. Internal view.
- c. External view.
- d. Physical View.

21- In the following code segment what will be the result of the function,

```
value of x , value of y
{unsigned int x=-1;
 int y;
 y = ~0;
 if(x == y)
 printf("same");
 else
 printf("not same");
 }
```

- a. same, MAXINT, -1
- b. not same, MAXINT, -MAXINT

- c. same , MAXUNIT, -1
 - d. (D) same, MAXUNIT, MAXUNIT
- 22- When applied to a variable, what does the unary `&l` operator yield?**
- a. Choice 1 The variable's value
 - b. Choice 2 The variable's binary form
 - c. Choice 3 The variable's address
 - d. Choice 4 The variable's format
- 23- What is a Firewall in Computer Network?**
- a. The physical boundary of Network
 - b. An operating System of Computer Network
 - c. A system designed to prevent unauthorized access
 - d. A web browsing Software
- 24- Which of these tags are all `<table>` tags?**
- a. `<table><head><tfoot>`
 - b. `<table><tr><td>`
 - c. `<table><tr><tt>`
 - d. `<thead><body><tr>`
- 25- How many layers does OSI Reference Model has?**
- a. 4
 - b. 5
 - c. 6
 - d. 7
- 26- DHCP is the abbreviation of**
- a. Dynamic Host Control Protocol
 - b. Dynamic Host Configuration Protocol
 - c. Dynamic Hyper Control Protocol
 - d. Dynamic Hyper Configuration Protocol
- 27- DML is provided for**
- a) Description of logical structure of database.
 - b) Addition of new structures in the database system.
 - c) Manipulation & processing of database.
 - d) Definition of physical structure of database system.
- 28- IPV4 Address is**
- a. 8 bit
 - b. 16 bit
 - c. 32 bit
 - d. 64 bit
- 29- DNS is the abbreviation of**
- a. Dynamic Name System
 - b. Dynamic Network System
 - c. Domain Name System
 - d. Domain Network Service

30- What is the meaning of Bandwidth in Network?

- a. Transmission capacity of a communication channels
- b. Connected Computers in the Network
- c. Class of IP used in Network
- d. None of Above

31- What is the use of Bridge in Network?

- a. to connect LANs
- b. to separate LANs
- c. to control Network Speed
- d. All of the above

32- Router operates in which layer of OSI Reference Model?

- a. Layer 1 (Physical Layer)
- b. Layer 3 (Network Layer)
- c. Layer 4 (Transport Layer)
- d. Layer 7 (Application Layer)

33- Each IP packet must contain

- a. Only Source address
- b. Only Destination address
- c. Source and Destination address
- d. Source or Destination address

34- Bridge works in which layer of the OSI model?

- a. Application layer
- b. Transport layer
- c. Network layer
- d. Datalink layer

35- _____ provides a connection-oriented reliable service for sending messages

- a. TCP
- b. IP
- c. UDP
- d. All of the above

36- Which layers of the OSI model are host-to-host layers?

- a. Transport, Session, Presentation, Application
- b. Network, Transport, Session, Presentation
- c. Datalink, Network, Transport, Session
- d. Physical, Datalink, Network, Transport

37- Which of the following IP address class is Multicast

- a. Class A
- b. Class B
- c. Class C
- d. Class D

38- Cartesian product in relational algebra is;

- a. a Unary operator.
- b. a Binary opera

- c. a Ternary operator.
- d. not defined.

39- _____ keyword is used to declare variables in JavaScript.

- a. Var
- b. Dim
- c. String
- d. None of the above

40- 'AS' clause is used in SQL for

- a) Selection operation.
- b) Rename operation.
- c) Join operation.
- d) Projection operation.

SECTION B: PROGRAMMING (20 MARKS)

1. Consider the following algorithm given below

```
1. start
2. variables: x=0, i,j,k,N as integer
3. for (i=0; i<N; i+1)
4.     for (j=0; j<N; j+1)
5.         for (k=0; k<N; k+1)
6.             x=x+1
7.             Display x
8.         endfor
9.     endfor
10. endfor
11. stop
```

Task:

- a) Give the equivalent flow chart of the algorithm
- b) Find the space complexity of the algorithm
- c) Find the time complexity of the algorithm
- d) What is the output of the algorithm if N=3

(10 marks)

(3 marks)

(3 marks)

(4 marks)

SECTION C: DATABASE**(15 MARKS)**

- 1) What are the commands used in DCL? (3 marks)
- 2) Explain the importance of database partitioning (2 marks)
- 3) What are the different authentication modes in SQL Server and how can you change authentication mode? (2 marks)
- 4) Differentiate between Atomicity and Aggregation. (3 marks)
- 5) Does the relational model, as seen by an SQL query writer, provide physical and logical data independence? Explain (3 marks)
- 6) What is Replication? (2 marks)

SECTION D: NETWORKING**(15 MARKS)**

- 1) Give the role of session Layer in an OSI model. (3 marks)
- 2) What is HTTPS and what port does it use? (3 marks)
- 3)
 - a) Which protocol requires a three-way handshake before transmission? (1 mark)
 - b) How does a three-way handshake function? (3 marks)
- 4)
 - a) Why are error detections necessary in networking? (2 marks)
 - b) Give three types of errors in networking. (3 marks)

SECTION E: DATA STRUCTURE AND ALGORITHM (10 MARKS)

- 1)
 - a) Draw a binary Tree for the expression: $(A+B)*C+(D-E)/F+G*A$. (2 marks)
 - b) Give the pre-order traversal of the binary tree. (2 marks)
 - a) Give the post-order traversal of the binary tree. (2 marks)
- 2) Give the types of participation constraints can you can find in an E-R model? (2 marks)
- 3) What are the three major steps of the database design (data modeling) process? Define each by one sentence. (2 marks)